Makanak Platform — Project Development Plan

Team Leader: Momen Ahmed
Project Duration: 4 Weeks
Project Type: Full-Stack .NET Real Estate Platform

Week 1: Requirements Analysis & System Design

Objective: Define the project scope, identify requirements, and prepare the system design before development begins.

Tasks

- Define the project vision, objectives, and scope.
- Define Functional and Non-Functional Requirements.
- Write User Stories and Use Cases.
- Design the UI/UX and user flows using Figma or Wireframes.
- Design the database schema and Entity Relationship Diagram (ERD).

Deliverables

- Project Scope Document
- User Stories and Use Cases
- UI/UX Designs and Wireframes
- Database Schema and ERD

Week 2: Backend Development

Objective: Implement the backend architecture, database, core business logic, and RESTful APIs.

Tasks

- Implement Clean Architecture and the Repository and Unit of Work patterns.
- Configure the database and Entity Framework Core.
- Implement Authentication and Authorization using JWT.
- Develop Property Management APIs (Create, Read, Update, and Delete).

- Implement property image upload and input validation.
- Develop property search, filtering, sorting, and pagination.
- Handle database transactions and concurrency conflicts.
- Implement centralized exception handling and API validation.
- Test and document the implemented endpoints.

Deliverables

- Configured .NET Backend Solution
- Database and Entity Relationships
- Authentication and Authorization
- Property Management APIs
- Property Search and Filtering APIs
- Input Validation and Error Handling
- API Documentation using Swagger/OpenAPI

Week 3: Frontend Development

Objective: Build the user interface and integrate it with the backend APIs.

Tasks

- Set up the frontend project structure and routing.
- Implement the main layout and responsive design.
- Develop Registration and Login pages.
- Integrate authentication and authorization with the backend.
- Build the homepage and property listing pages.
- Implement property search, filtering, sorting, and pagination interfaces.
- Develop property details pages with images and location information.
- Build property creation and editing forms for owners.
- Develop user dashboards for managing properties and offers.

- Implement the Offers and Counter Offers interfaces.
- Display offer statuses and negotiation history.
- Integrate all frontend features with the backend APIs.
- Handle loading states, validation errors, and API failures.
- Review the user experience and responsive layouts.

Deliverables

- Responsive User Interface
- Registration and Login Pages
- Homepage and Property Listings
- Property Search and Details Pages
- Owner Dashboard
- Property Management Forms
- Offers and Negotiation Interfaces
- Frontend–Backend Integration

Week 4: Testing, Code Review & Final Delivery

Objective: Verify system functionality, fix defects, review code quality, and prepare the final deliverables.

Tasks

- Test authentication, authorization, and access permissions.
- Validate property management and search functionality.
- Test offer submission, counter offers, acceptance, and rejection.
- Test concurrency handling and prevent conflicting offer acceptance.
- Perform end-to-end testing of the main user workflows.
- Test frontend responsiveness and usability.
- Review security, input validation, and error handling.

- Fix identified bugs and regressions.
- Conduct code reviews and verify project architecture.
- Finalize database and architecture documentation.
- Prepare the README and setup instructions.
- Configure the deployment environment and verify the production build.
- Prepare the final presentation and project demonstration.

Deliverables

- End-to-End Test Results
- Bug Fixes and Verified Workflows
- Final Code Review
- Updated README and Setup Instructions
- Database and Architecture Documentation
- Deployment-Ready Application
- Final Project Presentation and Demo
- Final Working MVP

Project Scope Notes

- The project will follow Clean Architecture and use the Repository and Unit of Work patterns.
- RESTful APIs will be the primary communication mechanism between the frontend and backend.
- SignalR and real-time notifications are excluded from the current MVP scope.
- Standard API requests will be used to retrieve updated offer statuses.
- Core functionality must be tested throughout development, not only during the final week.
- Additional features must be evaluated against the four-week deadline before being added to the scope.

Final Project Goal

Deliver a functional MVP of the Makanak Platform within four weeks, enabling users to discover properties, allowing owners to manage their listings, and supporting direct price negotiation through an offer and counter-offer workflow, with appropriate testing, documentation, and a final demonstration.